



NSW Education Standards Authority

2021 HIGHER SCHOOL CERTIFICATE EXAMINATION

Food Technology

**General
Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen

**Total marks:
100****Section I – 20 marks** (pages 2–6)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 50 marks (pages 9–16)

- Attempt Questions 21–26
- Allow about 1 hour and 25 minutes for this section

Section III – 15 marks (page 33)

- Attempt Question 27
- Allow about 30 minutes for this section

Section IV – 15 marks (page 34)

- Attempt Question 28
- Allow about 30 minutes for this section

Section I

20 marks

Attempt Questions 1–20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

- 1** What is the name of the group of additives used to replace sugar in food manufacturing?
- A. Emulsifiers
 - B. Preservatives
 - C. Flavour enhancers
 - D. Artificial sweeteners
- 2** A food manufacturer introduces a new packaging size for an existing product they produce.
- This product is known as a
- A. me-too.
 - B. copy-cat.
 - C. line-extension.
 - D. new-to-the-world.
- 3** The main purpose of food dehydration in food manufacture is to
- A. rehydrate food.
 - B. prolong the shelf life of food.
 - C. concentrate the flavour of the food.
 - D. reduce the size for ease of packaging.
- 4** An individual who does not eat any food derived from animals or animal-sourced products is most likely to be considered
- A. vegan.
 - B. vegetarian.
 - C. pescatarian.
 - D. lactose intolerant.

- 5 Which of the following is a breach of the *Work Health and Safety Act 2011* (Cth)?
- A. Serving expired food
 - B. Not reporting faulty equipment
 - C. Not declaring allergens on food labels
 - D. Providing inaccurate weight on food packaging
- 6 Which of the following may be a consequence of consuming a high salt diet?
- A. Obesity
 - B. Dental caries
 - C. Hypertension
 - D. Vitamin deficiency
- 7 Industrial food production equipment is required to be
- A. durable, fast and efficient.
 - B. precise, consistent and large.
 - C. sturdy, manually operated and large.
 - D. durable, manually operated and efficient.
- 8 Which sector of the Australian food industry would get the greatest benefit from an increase in people using apps to order take-away food?
- A. Food retail
 - B. Agriculture and fisheries
 - C. Food service and catering
 - D. Food manufacture and processing
- 9 A person working in food retail is successful in obtaining a scholarship for a course in management training. This is an example of
- A. an employee right.
 - B. a career opportunity.
 - C. enterprise bargaining.
 - D. improving company image.

- 10** Which set of factors is required to successfully market a new food product?
- A. Popularity, planning, product and price
 - B. Product, product features, price and placement
 - C. Place, price, product and promotional program
 - D. Planning, promotional program, place and distribution system
- 11** A fast food company is sponsoring a major sporting event. This is an example of
- A. direct selling.
 - B. cross promotion.
 - C. consumer testing.
 - D. competition give-away.
- 12** A child complains of tingling in the mouth, swelling of the face and feeling dizzy.
These symptoms are most likely to be caused by
- A. a food allergy.
 - B. a diabetic reaction.
 - C. a diet high in fibre.
 - D. an excess vitamin intake.
- 13** The primary role of advisory groups in the Australian food industry is to
- A. market products to consumers.
 - B. place tariffs on new food products.
 - C. develop new food products for a food manufacturer.
 - D. provide guidance to government when developing policies.
- 14** A food manufacturer is required to pay a tax on an imported ingredient. Which of the following is most likely to be responsible for this tax?
- A. Political decision
 - B. Company profitability
 - C. Foreign trade partnerships
 - D. Technological advancements

15 Food manufacturers have increased their use of biodegradable packaging materials.

Which of the following has enabled this change?

- A. Personnel expertise
- B. Market concerns for health
- C. Technological developments
- D. Consumer demands for convenience

16 During which step of the food product development process would a food manufacturer determine if the product is likely to be a financial success?

- A. Market research
- B. Feasibility study
- C. Product specifications
- D. Development of product prototype

17 Which of the following is the best example of a functional, fortified food?

- A. Custard
- B. Sultanas
- C. Cake mix
- D. White bread

18 Lack of available fruit pickers has left Australian farmers with unpicked, rotting fruit.

Which of the following is most likely to be affected?

- A. The economy
- B. Quality assurance
- C. Working conditions
- D. Research and development

- 19** A new yoghurt has been released which has added fibre and probiotics. These ingredients are
- A. laxatives.
 - B. food additives.
 - C. diet supplements.
 - D. active non-nutrients.

- 20** A supermarket has introduced data embedded barcodes to their meat and poultry labels to help manage product recalls, food waste and sales of expired food products.

What is this an example of?

- A. Active packaging
- B. A preservation process
- C. A current development in technology
- D. A legislated food labelling requirement

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Centre Number

Food Technology

Sections II, III and IV

Answer Booklet

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Student Number

Instructions

- Write your Centre Number and Student Number at the top of this page.
- Use this answer booklet for Sections II, III and IV.
- Follow the instructions inside this booklet as to where you should write your answers.

Please turn over

Section II

50 marks

Attempt Questions 21–26

Allow about 1 hour and 25 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Question 21 (6 marks)

A food manufacturer wants to release a *me-too* version of an existing popular sandwich spread.

- (a) Outline the roles of BOTH idea generation and screening in the development of this product. 2

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- (b) Conduct a SWOT analysis for this *me-too* version. 4

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Question 22 (6 marks)

- (a) Outline ONE emerging technology in the Australian food industry. **2**

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- (b) Describe potential benefits of an emerging technology in ONE sector of the Australian food industry. **4**

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Question 23 (9 marks)

- (a) Outline the features required in equipment to be used by a household company in the production of a food product. **2**

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- (b) Describe the specifications food manufacturers need to consider when selecting raw materials. **3**

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- (c) Propose possible waste management strategies a food manufacturing company could implement. **4**

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Question 24 (13 marks)

- (a) Outline the causes of food deterioration that affect the shelf life of cut fruit and vegetables. **3**

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- (b) Recommend a preservation process that could be used to prolong the shelf life of a named fruit or vegetable. **4**

Fruit or vegetable:

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Question 24 continues on page 14

Question 24 (continued)

- (c) Explain how food product developers have responded to consumer demands for healthier products. Support your response with examples.

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End of Question 24

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Question 25 (10 marks)

- (a) How have food product developers responded to the nutritional considerations for a specific group in Australia? **3**

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- (b) What is the role of ONE organisation in promoting health in Australia? **3**

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- (c) Explain the physical consequences of malnutrition. **4**

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Question 26 (6 marks)

How does ONE organisation within the Australian food industry provide consumers with quality assurance? Support your response with examples.

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Section III — Answer Question 27 on pages 17–23

27

Question Number

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If you require more space to answer Section III, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Section III, tick here.

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Section IV — Answer Question 28 on pages 24–30

28

Question Number

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If you have used an extra writing booklet for Section IV, tick here.

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Food Technology

Section III

15 marks

Attempt Question 27

Allow about 30 minutes for this section

Answer the question on pages 17–23 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Question 27 (15 marks)

An Australian fruit juice manufacturer has developed a new range of fruit drinks with 50% less sugar and added dietary fibre.

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|-----|---|----------|
| (a) | Outline TWO pieces of legislation that would need to be followed by the food manufacturer during the development of this product. | 3 |
| (b) | Describe ONE test that could be used to determine the suitability of a proposed package for the newly developed juice. | 4 |
| (c) | Explain how this nutritionally modified juice can improve the health of Australians. | 8 |

Please turn over

Section IV

15 marks

Attempt Question 28

Allow about 30 minutes for this section

Answer the question on pages 24–30 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Your answer will be assessed on how well you:

- demonstrate knowledge and understanding relevant to the question
 - apply course concepts to food technology issues
 - communicate ideas and information using appropriate terminology and relevant examples
 - present a logical and cohesive response
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Question 28 (15 marks)

Evaluate how sectors of the Australian food industry have responded to BOTH societal changes and consumer demands for convenience.

End of paper